

# Timothy Do

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## EDUCATION

### University of California, Los Angeles

Sep. 2023 – Jun. 2024

*M.S. Electrical and Computer Engineering, Signal Processing Specialization*

*Los Angeles, CA*

- **Cumulative GPA:** 3.933/4.0
- **Relevant Coursework:** Computational Imaging, Digital Speech Processing, Advanced Deep Learning & Neural Networks, Modern Wireless Communication Systems, Large-Scale Social Complex Networks
- **Clubs and Societies:** ECEGAPS (Electrical and Computer Engineering Graduate and PostDoc Society)

### University of California, Irvine

Sep. 2019 – Jun. 2023

*B.S. Electrical Engineering, Digital Signal Processing & Communications Specialization*

*Irvine, CA*

- **Cumulative GPA:** 3.924/4.0, **Major GPA:** 3.944/4.0, **Magna Cum Laude, Accelerated Status**
- **Relevant Coursework:** Computer Architecture, Control Systems, Digital Image/Signal Processing, Electronics
- **Clubs and Societies:** Irvine Computer Vision Laboratory, IEEE, IPC, Eta Kappa Nu, Tau Beta Pi

## EXPERIENCE

### Computer Vision Software Engineer

Nov. 2024 - Present

*SiMa.ai*

*San Jose, CA*

- Auto-generated documentation for SiMa.ai's product family and created technical videos on how to build applications on the Palette SDK.
- Led GStreamer-based projects for OCR, Change Detection, and Visual Language Models, managing the full pipeline from data collection and model finetuning to quantization and deployment.
- Supported customers end to end, from the optical selection, imaging process, to perform factory acceptance testing and validation through 50+ product lines running on SiMa.ai accelerator.

### Computer Vision Intern

Aug. 2024 - Nov. 2024

*SiMa.ai*

*San Jose, CA*

- Deployed and presented a production-level Anomaly Detection demo on SiMa.ai's MLSoC at the 2024 AI Hardware & Edge AI Summit within two weeks of joining.
- Led project tasks including data collection, model finetuning, and deployment for Anomaly and Change Detection, while pitching technical solutions to industry leaders.

### Embedded Software Engineering Co-Op

Jun. 2023 – Apr. 2024

*Maxeon Solar Technologies*

*San Jose, CA*

- Implemented Robust OpenCV pipelines to quantitatively detect module features for defect binning
- Developed Tkinter GUI interfaces for operator ease of use when testing images under custom data pipelines
- Trained robust YOLOv5/YOLOv8 Module Defect Classification Models deployed in manufacturing fab

### Lead Undergraduate Student Researcher

Jan. 2022 – Jun. 2023

*Irvine Computer Vision Laboratory at UC Irvine*

*Irvine, CA*

- Researched Spectral Image Filtering and Baseball Simulations under Professor Glenn Healey
- Digitized Hyperspectral Optical Imaging Filters into CSVs and compared them against certain chemical emissions
- Managed 15+ Graduate/Undergraduate Researchers in Google Spaces and developed shared MATLAB/Python Scripts hosted on Google Drive/Github for analyzing SWIR/NVIR vegetation spectras

### Industrial Internet of Things Intern

Jun. 2022 – Sep. 2022

*Western Digital*

*San Jose, CA*

- Applied Industrial Toolsets to develop a Realtime Raspberry Pi Sensor Data Analytics Platform
- Earned Six Sigma Yellow Belt award for learning foundations of Lean Six Sigma through LinkedIn Learning
- Saved the company an annual projected cost of \$312,000 from hazard mitigations in the manufacturing fab

## PROJECTS

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**End-to-End Automatic Modulation Recognition** | *Python, MATLAB, Pytorch* Jan. 2024 – Mar. 2024

- Designed a CNN-based AMR System on I/Q samples (93.2% test accuracy at high SNR)
- Preprocessed multiple modulation datasets by one-hot encoding integer labels for minimizing cross-entropy loss
- Applied data augmentation by index reversal on each of the data samples for a 3% performance improvement

**African-American Dialect Classification** | *Python, Librosa, Spafe, Opensmile* Feb. 2024 – Mar. 2024

- Classified different dialects of African Americans across the east coast with the COORAL dataset
- Developed a XGBoost Classifier using Opensmile and GFCC features for 95% test accuracy
- Applied chatter noise augmentation to the samples resulting in improving accuracy by 2%
- Implemented multiprocessing pipeline to extract multiple samples simultaneously (3-4X speedup)

**EEG Action Classification** | *Python, Tensorflow, Pytorch* Feb. 2024 – Mar. 2024

- Explored various model architectures on EEGs to classify a subject's actions, the best yielding 75.4% test accuracy
- Performed Data Cleaning on the BCI dataset by trimming and maxpooling to improve classification accuracy
- Optimized model hyperparameters by automating exhaustive grid search on learning rate, batch size, and epochs

## SKILLS

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**Electronics:** Raspberry Pi, Network Analysis, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements

**Programming Languages:** Python, Java, C, CUDA, MATLAB, bash, MySQL HTML/CSS/Javascript, VHDL

**Developer Tools:** git, SSH, Makefile, Linux, Azure, VS Code, Visual Studio, PyCharm, Eclipse, Vivado HLS

**Libraries:** OpenCV, Pandas, Numpy, Matplotlib, PIL, Tensorflow, Pytorch, Tkinter, Socket, Flask, Librosa